Chapter 1: Transport Layer Protocols

TCP (Transmission Control Protocol) is a connection-oriented protocol.
It provides reliable, ordered, and error-checked delivery of a stream of bytes between applications running on ho
Major applications that require TCP include the World Wide Web, email, and file transfer.

Chapter 2: Connectionless Protocols

UDP (User Datagram Protocol) is a connectionless transport protocol.
Unlike TCP, UDP does not guarantee packet delivery, ordering, or duplicate protection.
It is widely used in time-sensitive applications such as video streaming, VoIP, DNS queries, and online gaming w

Chapter 3: Application Layer & HTTP

HTTP (Hypertext Transfer Protocol) is an application-layer protocol for transmitting hypermedia documents.
HTTP is stateless, meaning that each command is executed independently, without any knowledge of the comm
HTTP commonly uses TCP port 80, while HTTPS uses port 443.